CS24303 — THEORY (3.0 CR) | 10-PAGE MASTER REVISION | BIT MESRA | NEP SCHEME

Data Mining — 10-Page Master Quick Revision

High-Yield Formula Sheet, Schema Matrices & Top BIT Mesra PYQ Solutions

⛏ 10-Page Master Quick Revision — Data Mining Concepts & Techniques (CS24303)

⚡ Master Formula, Schema & Metric Matrix

Core Concept	Mathematical Formulation / Rule	Key Exam Insight
Euclidean Distance (L_2)	$d(x,y) = \sqrt{\sum_{i=1}^{p}(x_i - y_i)^2}$	Standard straight-line distance; sensitive to scale.
Manhattan Distance (L_1)	$d(x,y) = \sum_{i=1}^{p} \lvert x_i - y_i \rvert$	City-block grid distance; robust to single-attribute outliers.
Cosine Similarity	$\mathrm{sim}(x,y) = \frac{\vec{x}\cdot\vec{y}}{\lVert\vec{x}\rVert\lVert\vec{y}\rVert}$	Invariant to document vector length.
Jaccard Coefficient	$J(A,B) = \frac{q}{q+r+s}$	Asymmetric binary; ignores joint absence (t).
Chi-Square Test (χ^2)	$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad E_{ij} = \frac{\mathrm{Row}_i \times \mathrm{Col}_j}{N}$	Tests independence between two nominal attributes.
Min-Max Normalization	$v' = \frac{v - \min_A}{\max_A - \min_A}(new\_\max - new\_\min) + new\_\min$	Linearly maps to target interval $[0,1]$.
Z-Score Normalization	$v' = \frac{v - \mu_A}{\sigma_A}$	Zero mean, unit variance. Robust against outliers.
Rule Support & Confidence	$\mathrm{Supp} = \frac{\sigma(A \cup B)}{\lvert D \rvert}, \qquad \mathrm{Conf} = \frac{\sigma(A \cup B)}{\sigma(A)}$	Supp measures frequency; Conf measures rule certainty.
Lift Metric	$\mathrm{Lift}(A,B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{\mathrm{Conf}(A \implies B)}{\mathrm{Supp}(B)}$	> 1: Positive correlation; < 1 : Negative correlation.
Kulczynski Measure	$\mathrm{Kulc}(A,B) = \frac{1}{2}\Big(P(A \mid B) + P(B \mid A)\Big)$	Null-invariant correlation measure.

🔥 Top 10 High-Yield BIT Mesra Exam Questions & Model Answers

Q1. Explain the Apriori property and how it reduces candidate itemset generation. (8 Marks)

Apriori Property (Downward Closure): All non-empty subsets of a frequent itemset must also be frequent. If an itemset I is infrequent ($\mathrm{count}(I) < \mathrm{min\_sup}$), any superset containing I cannot be frequent.
Pruning Mechanism: When generating candidate k-itemsets (C_k) by joining $L_{k-1} \bowtie L_{k-1}$, if any $(k-1)$-subset of a candidate $c \in C_k$ is not in L_{k-1}, c is immediately pruned from C_k without scanning the database.

Q2. Compare Star Schema and Snowflake Schema across 5 technical criteria. (8 Marks)

1. **Structure:** Star has single central fact table and denormalized dimension tables; Snowflake normalizes dimension tables into multiple sub-tables.
2. **Redundancy:** Star contains data redundancy in dimensions; Snowflake eliminates redundancy.
3. **Storage:** Star requires slightly more disk space; Snowflake is storage-optimized.
4. **Query Performance:** Star requires simple single-join SQL queries with superior performance; Snowflake requires complex multi-table joins reducing query speed.
5. **Maintenance:** Star has simpler ETL updates; Snowflake has complex hierarchical maintenance.

Q3. Detail the difference between ROLAP, MOLAP, and HOLAP servers. (6 Marks)

ROLAP: Relational OLAP storing data in RDBMS with Star/Snowflake schemas. Scalable to petabytes, leverages SQL.
MOLAP: Multidimensional OLAP storing data in dense arrays. Blazing fast query times, pre-computed materialization.
HOLAP: Hybrid OLAP storing base records in ROLAP relational tables and high-level aggregated summary cuboids in MOLAP arrays.

Q4. State the difference between Closed Frequent Itemsets and Maximal Frequent Itemsets. (6 Marks)

Closed Frequent Itemset: An itemset X is closed if no proper superset $Y \supset X$ has the same support count ($\text{supp}(Y) = \text{supp}(X)$). It is a *lossless representation* (exact counts of all subsets can be derived).
Maximal Frequent Itemset (Max-Itemset): An itemset X is maximal frequent if no proper superset $Y \supset X$ is frequent. It is a *lossy representation* (subset counts are not uniquely recoverable).

Q5. Explain the two core operators in Attribute-Oriented Induction (AOI). (6 Marks)

1. Attribute Removal: If an attribute contains a large number of distinct values with no concept hierarchy (e.g., `Social_Security_Number`), remove it from the generalized relation.
2. Attribute Generalization: If an attribute has a concept hierarchy and distinct values exceed attribute threshold, replace lower-level values with higher-level concepts.

Q6. Differentiate between Descriptive and Predictive Data Mining with examples. (6 Marks)

Descriptive Mining: Characterizes general structural properties of existing data (Clustering, Association Rules, Data Summarization). Example: Segmenting customers into 4 distinct demographic spending clusters.
Predictive Mining: Constructs inference models based on historical labeled data to forecast unknown or future values (Classification, Regression). Example: Predicting whether a loan applicant will default based on credit score.

Q7. What is the difference between Interdimension and Hybrid Association Rules? (5 Marks)

Interdimension Rules: Contain multiple distinct predicates where each predicate occurs at most once (e.g., $\text{Age}(X, 20) \wedge \text{Income}(X, \text{High}) \implies \text{Buys}(X, \text{Laptop})$).
Hybrid Rules: Contain multiple predicates where one or more predicates may occur repeatedly with different attribute values (e.g., $\text{Age}(X, 20) \wedge \text{Buys}(X, \text{Laptop}) \implies \text{Buys}(X, \text{Backpack})$).